Or $\left|5-\frac{2}{n}\right| < 1$

$$\left(5-\frac{2}{n}\right) < \pm 1$$

$$\left(5-\frac{2}{n}\right) < +1$$

$$\left|5-\frac{2}{x}\right|^2 < (1)^2$$

$a^2 - 2ab + b^2$

$$\left(5-\frac{2}{x}\right)^2 < 1$$

$$5^2 - 2(5)\left(\frac{2}{x}\right) + \left(\frac{2}{x}\right)^2 < 1$$

$$25 - \frac{20}{x} + \frac{4}{x^2} < 1$$

Q2 $\left|5-\frac{2}{x}\right| < 1$

$\left(5-\frac{2}{x}\right) < \pm 1$

$\left(5-\frac{2}{x}\right) < +1, \left(5-\frac{2}{x}\right) < -1$

$+\frac{2}{x} < +4 \quad , \quad +\frac{2}{x} < +6$

$2 < 4x \quad , \quad 2 < 6x$

$1/2 < x \quad , \quad 1/3 < x$

$(1/2, \infty) \cup (-\infty, -1/3)$

Q3 $|x+1| \geq 6$

$(x+1) \geq \pm 6$

$(x+1) \geq +6 \quad , \quad (x+1) \geq -6$

$x \geq 5 \quad , \quad x \geq -7$

$[5, \infty) \cup [-\infty, 7]$

Sets:

orderly arrangement of well defined objects.

Common sets:

1. Set of natural numbers:

   counting numbers.

   Example: $\{1, 2, 3, 4, \ldots\}$

2. Set of whole numbers:

   Natural numbers + zero.

   Example: $\{0, 1, 2, 3 \ldots\}$

3. Set of integers:

   whole numbers and their negatives.

   Example: $\{\ldots -3, -2, -1, 0, 1, 2, 3 \ldots\}$

4. Set of rational numbers:

   Numbers that can be written as a fraction p/q, where $q \neq 0$.

   Example: $\{1/2, 3/4, -5/6, 2, 0\}$

Set types:

1. Empty (Null) set:
   Set with no elements.
   Example: $\varnothing$ or $\{\ \}$

2. Finite set:
   A set with a limited number of elements.
   Example: $\{1, 2, 3, 4\}$

3. Infinite set:
   A set with infinitely many elements.
   Example: $\mathbb{N} = \{1, 2, 3 \ldots\}$.

4. Equal sets:
   Sets having exactly the same elements.
   Example: $\{1, 2, 3\} = \{3, 2, 1\}$

5. Subset:
   All elements of one set are contained in another
   Example: $A = \{1, 2\}, B = \{1, 2, 3\} \rightarrow A \subseteq B$.

Set operations:

① Union ($\cup$):

All elements that are in either set (or both)

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

② Intersection ($\cap$):

Elements that are common to both sets.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

③ Difference ($-$):

Elements in one set but not in the other

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

Function:

Special type of relation.

Relation → One input → Many output

Function → One input → One output

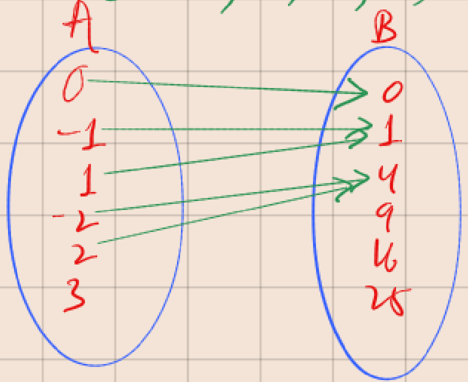
$f(x)$ input , $f: A \to B$ → Co-domain

domain.

$f(x) = x^2$

Domain : 0, -1, 1, -2, 2, 3.

Co-domain : 0, 1, 4, 9, 16, 25.



Absolute value function:

$|x|$ → +, −

$(?, ?)$] output (+) ?

distance

vertex (0,0)

symmetry about y-axis → Domain $(-\infty, \infty)$, $[0, \infty)$

for vertex

$a|x-h|+k$

(h,k)
vertex

$a \quad 2|x+2|+1$

$h \quad k$

vertex $= (-2, 1)$

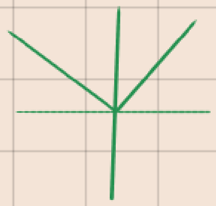
$x+2=0$

$x=-2$

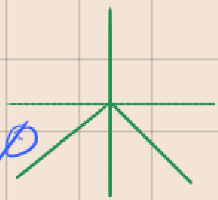
$f(x)=1$

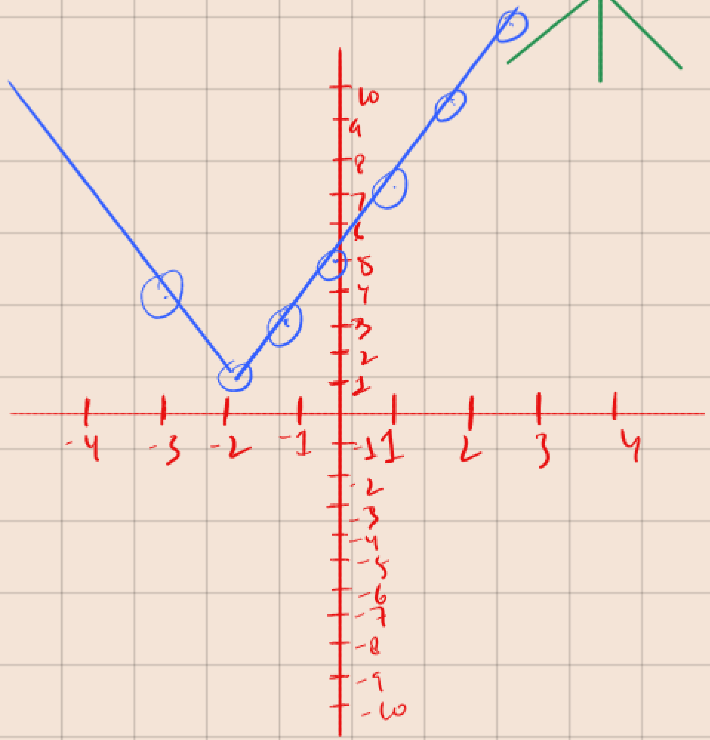
$(-2, 1)$

$a>0$



$a<0$





x	$f(x)$
0	5
1	7
-1	3
2	9
-2	1
3	11
-3	3